A

Project Report

on

FAKE ACCOUNT DETECTION USING MACHINE LEARNING AND DATA SCIENCE

Submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology

in

COMPUTER SCIENCE & ENGINEERING (DATA SCIENCE)

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April, 2025

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (DATA SCIENCE)

CERTIFICATE

This is to certify that the project report entitled **"Fake Account Detection Using Machine Learning And Data Science"** being submitted by **E.PAVANKUMAR (217R1A67E2), CH.SANJAYKUMAR (217R1A67D7), R.KARTHIK (217R5A6716)** in partial fulfillment of the requirements for the award of the degree of B.Tech in **Computer Science and Engineering (Data Science)** to the Jawaharlal Nehru Technological University Hyderabad, is a record of bonafide work carried out by them under our guidance and supervision during the Academic Year 2024-2025.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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Assistant Professor
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External Examiner

Submitted for viva-voce Examination held on ________________

ACKNOWLEDGMENT

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ABSTRACT

In today's digital age, online social media has become a dominant force, revolutionizing the way people connect and communicate. With the number of users growing exponentially, these platforms offer unparalleled convenience in staying connected. However, this rapid expansion has also introduced significant risks, including the rise of fake identities and the spread of false information. Recent studies reveal that the number of social media accounts far exceeds the actual user base, indicating a sharp increase in fake profiles. These fraudulent accounts pose a major challenge for social media providers, as they struggle to detect and mitigate them effectively. The proliferation of fake accounts has led to an influx of misinformation, spam, and deceptive advertisements, undermining the integrity of these platforms.

Traditional detection methods, such as Naïve Bayes, Support Vector Machines (SVM), and Random Forest, have proven ineffective against increasingly sophisticated fake account creation techniques. Modern approaches now focus on identifying automated behaviors, such as spam comments, artificial activity patterns, and irregular engagement rates. In this research, we propose an innovative solution by leveraging the Gradient Boosting algorithm with Decision Trees, incorporating three key attributes—spam commenting, artificial activity, and engagement rate—to enhance detection accuracy. By combining Machine Learning and Data Science techniques, our model provides a more reliable and efficient way to identify fake accounts, addressing a critical challenge in today's social media landscape.

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Chapter 1

INTRODUCTION

1.1 Introduction

In today's Modern society, social media plays a vital role in everyone's life. The general purpose of social media is to keep in touch with friends, sharing news, etc. The number of users in social media is increasing exponentially. Instagram has recently gained immense popularity among social media users. With more than 1 Billion active users, Instagram has become one of the most used social media sites. After the emergence of Instagram to the social media scenario, people with a good number of followers have been called Social Media Influencers. These social media influencers have now become a go-to place for the business organization to advertise their products and services

The widespread use of social media has become both a boon and a bane for the society. Using Social media for online fraud, spreading False information is increasing at a rapid pace. Fake accounts are the major source of false information on social media. Business organizations that invest huge Sum of money on social media influencers must know whether the following gained by that account is organic or not. So, there is a widespread need for a fake account detection tool, which can accurately say whether the account is fake or not. In this paper, we use classification algorithms in machine learning to detect fake accounts. The process of finding a fake account mainly depends on factors such as engagement rate and artificial activity

1.2 Project Scope

With the increasing adoption of cloud computing, ensuring robust security measures has become a critical priority for organizations. The dynamic nature of cloud environments introduces evolving threats, making it essential to evaluate and quantify risks systematically. This project aims to develop a structured approach for assessing cloud security threats using quantitative methods, enabling organizations to prioritize mitigation strategies effectively..

1.3 Project Scope

1.4 Project Features

Chapter 2
SYSTEM ANALYSIS

2.1 Problem Definition

Online social media platforms have become integral to modern communication, enabling users to connect, share information, and engage with content. However, the exponential growth of social media has also led to a surge in fake accounts, which are used for spreading misinformation, spam, fraudulent advertisements, and other malicious activities. Recent studies indicate that the number of fake accounts on platforms like Instagram, Facebook, and Twitter far exceeds the actual user base, posing significant challenges for platform providers and legitimate users.

2.2 Existing System

The existing systems use very fewer factors to decide whether an account is fake or not. The factors largely affect the way decision making occurs. When the number of factors is low, the accuracy of the decision making is reduced significantly. There is an exceptional improvement in fake account creation, which is unmatched by the software or application used to detect the fake account. Due to the advancement in creation of fake account, existing methods have turned obsolete. The most common algorithm used by fake account detection Applications is the Random forest algorithm. The algorithm has few downsides such as inefficiency to handle the categorical variables which has different number of levels. Also, when there is an increase in the number of trees, the algorithm's time efficiency takes a hit.

2.2.1 Limitations of The Existing System

Reliance on Outdated Algorithms
Dependency on Static Features
Inability to Handle Behavioral Dynamics
Poor Scalability
High False Positives/Negatives
Lack of Real-Time Processing
Limited Feature Engineering

2.3 Proposed System

The existing system uses random forest algorithm to identify the fake account. It is efficient when it has the correct inputs and when it has all the inputs. When some of the inputs are missing it becomes difficult for the algorithm to produce the output. To overcome such difficulties in the proposed systems we used a gradient boosting algorithm. Gradient boosting algorithm is like random forest algorithm which uses decision trees as its main component. We also changed the way we find the fake accounts i.e., we introduced new methods to find the account. The methods used are spam commenting, engagement rate and artificial activity. These inputs are used to form decision trees that are used in the gradient boosting algorithm. This algorithm gives us an output even if some inputs are missing. This is the major reason for choosing this algorithm. Due to the use of this algorithm we were able to get highly accurate results.

2.3.1 Advantages of Proposed System

Using Robust Features

Handling Missing Data

Real-Time Adaptability

Higher Accuracy

2.4 Feasibility Study

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

2.4.1 Economical Feasibility

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

2.4.2 Technical Feasibility

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

2.4.3 Social Feasibility

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

2.5 Hardware & Software Requirements

2.5.1 Hardware Requirements

- System : MINIMUM i3.
- Hard Disk : 40 GB.
- Ram : 4 GB.

2.5.2 Software Requirements

- Operating System: Windows 8
- Coding Language: Python 3.7

2.6 Programming Language(S) Used

The programming languages and software tools utilized in the development of the **Fake Account Detection Using Machine Learning and Data Science** project.

2.6.1. Java /Python/Any Other Front End

The front-end of the project was developed using **Python**. Python was chosen for its simplicity, extensive libraries, and seamless integration with machine learning frameworks.

Some of the key Python libraries used include:

- **Flask / Django**: For web-based front-end interface development.
- **Streamlit**: For building interactive dashboards.
- **Tkinter**: For GUI-based applications.
- **Matplotlib & Seaborn**: For visualization of data.
- **NumPy & Pandas**: For data manipulation and preprocessing.

If **Java** was used, it may have been utilized for front-end integration with **JavaFX** or **Spring Boot** for web-based front-end development.

2.6.3 SQL/Any Other Back End

For the backend, **SQL** databases were used to store and retrieve data efficiently. The database technologies include:

- **MySQL / PostgreSQL**: Used for structured data storage.
- **SQLite**: Used for lightweight, local storage.
- **MongoDB**: NoSQL alternative for handling unstructured data.

These databases were integrated with Python using **SQLAlchemy**, **PyMySQL**, and **MongoDB connectors**.

2.6.3. Any Other Software

In addition to programming languages, various other software tools were used to enhance the functionality and efficiency of the project, such as:

- **Jupyter Notebook / Google Colab**: For coding, testing, and data analysis.
- **Scikit-learn**: For implementing machine learning models.
- **TensorFlow / PyTorch**: For deep learning approaches.
- **Apache Spark**: For big data processing.

2.6.4. Any Other Software

Additional software tools and platforms used in the project include:

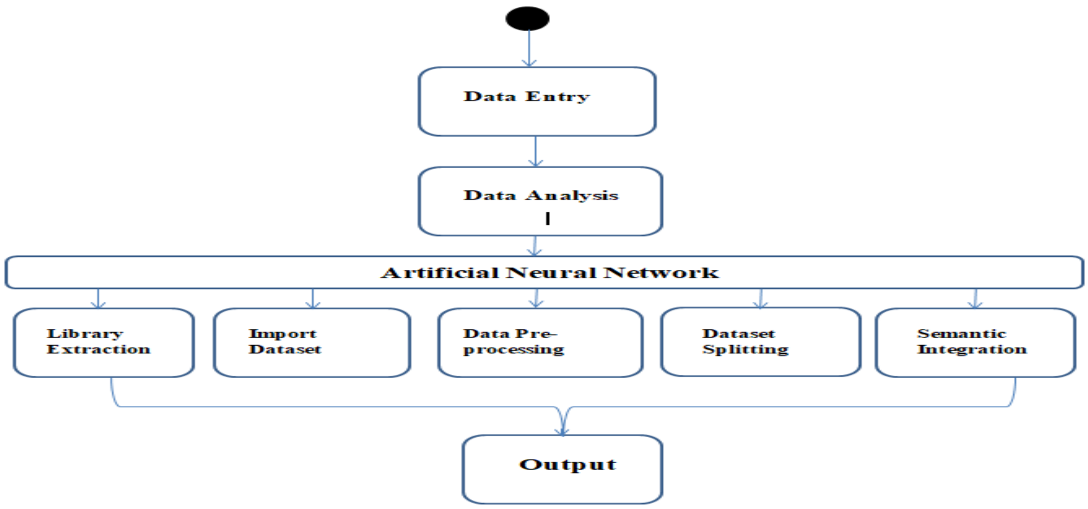
- **Git & GitHub**: For version control and collaborative development.
- **Docker**: For containerized deployment.
- **AWS / Google Cloud / Azure**: For cloud-based deployment and storage.
- **Jenkins**: For continuous integration and deployment.

These tools contributed to the seamless execution of the project by ensuring data security, efficient processing, and scalability.

Chapter 3

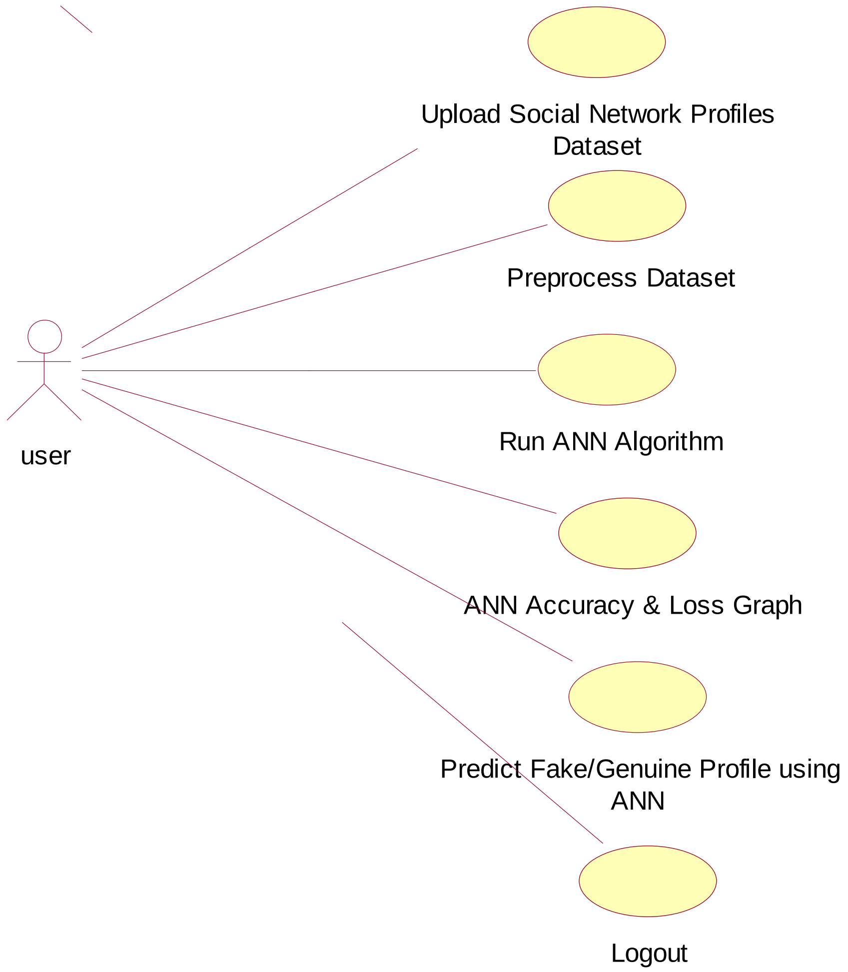
ARCHITECTURE

3.1 Project Architecture



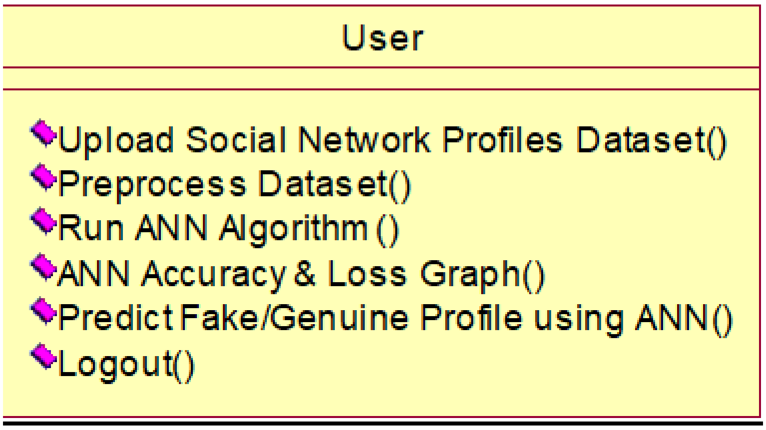
3.2 Use Case Diagram

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.



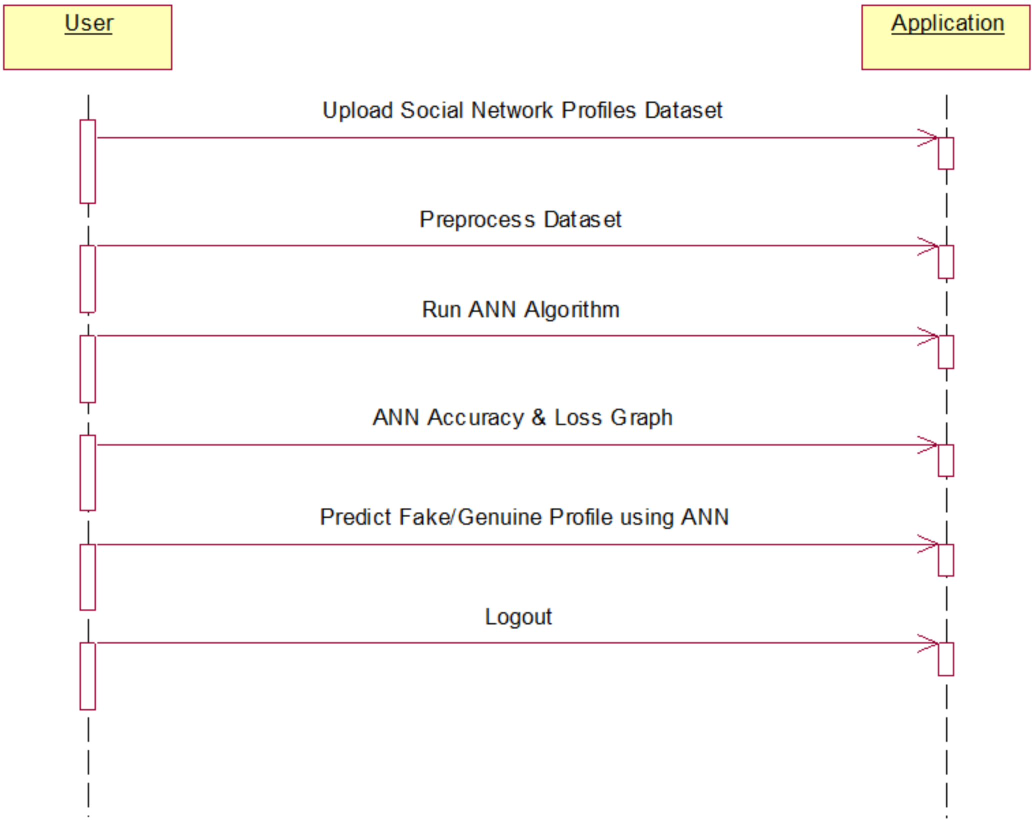
3.3 Class Diagram

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.



3.4 Sequence Diagram

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.



3.5 Activity Diagram

The **Activity Diagram** represents the workflow involved in detecting fake accounts, focusing on the interactions between two primary entities:

3.5.1 Remote User

A **Remote User** is an external entity that interacts with the system. This could be a normal user attempting to access services or a potential fake account. The key activities of a Remote User may include:

- **Account Registration/Login** – The user enters credentials or registers a new account.
- **Profile and Activity Submission** – The user engages in activities like posting content, sending messages, or updating profile details.
- **System Interaction** – Based on the user's behavior, the system evaluates authenticity and detects potential fake accounts.

3.5.2 Service Provider

The **Service Provider** represents the system responsible for handling user requests and ensuring fake account detection. Its key activities include:

- **Data Collection & Preprocessing** – The system gathers and processes user information (e.g., login patterns, activity logs, and profile details).
- **Machine Learning-Based Detection** – Algorithms analyze user behavior, comparing it against known patterns of fake and legitimate accounts.
- **Verification & Response** – If suspicious activity is detected, the system may trigger security measures like CAPTCHA verification, account suspension, or manual review.

Chapter 4

IMPLEMENTATION

4.1 Machine Learning Algorithms

Supervised Learning Algorithms

These algorithms are trained on labeled datasets (real vs. fake accounts) and make predictions based on features such as activity patterns, friend connections, and posting behavior.

- **Logistic Regression** – A statistical model used for binary classification (fake or real account).
- **Decision Trees** – A rule-based approach to classify users based on their behavioral characteristics.
- **Random Forest** – An ensemble method that improves accuracy by combining multiple decision trees.
- **Support Vector Machine (SVM)** – Helps in classifying fake and real accounts by finding an optimal decision boundary.

Unsupervised Learning Algorithms

These are used when labeled data is unavailable. They detect anomalies in user behavior that may indicate fake accounts.

- **K-Means Clustering** – Groups similar user accounts and detects outliers (suspicious users).
- **DBSCAN (Density-Based Spatial Clustering)** – Identifies clusters of real users and isolates fake ones.

Deep Learning Approaches

- **Neural Networks** – More advanced models like Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs) can analyze text patterns, images, or time-based behaviors.

Feature Engineering in Fake Account Detection

The system likely extracts key features such as:

- Account age
- Number of connections
- Activity frequency
- Posting patterns
- Profile completeness

4.2 Data Set Description

Introduction

The dataset is used for detecting fake accounts using machine learning and data science techniques. It contains a mix of user profile information and behavioral data to distinguish between real and fake accounts.

Source of Data: The dataset could be sourced from social media platforms, scraped publicly available data, or provided by research datasets.

Nature of Data:

- **Type**: Structured, tabular data
- **Size**: Could vary based on the number of accounts analyzed
- **Format**: Typically CSV, JSON, or database records

Characteristics of Data:

- The dataset includes structured data with numerical and categorical attributes.
- Features may include user activity metrics, profile completeness, engagement patterns, and other relevant indicators.

Variables and Attributes:

- **User ID** (Unique identifier)
- **Account Age** (Days since account creation)
- **Followers/Following Count** (Number of followers and accounts followed)
- **Post Frequency** (Activity level)
- **Engagement Rate** (Likes, comments, shares)
- **Profile Completeness** (Availability of profile picture, bio, etc.)
- **Verified Status** (Binary indicator)

4.3 Sample Code

Main.py

```
from tkinter import messagebox
from tkinter import *
from tkinter import simpledialog
import tkinter
import matplotlib.pyplot as plt
import numpy as np
from tkinter import ttk
from tkinter import filedialog
import pandas as pd
from sklearn.model_selection import train_test_split
from keras.models import Sequential
from keras.layers.core import Dense,Activation,Dropout
from keras.callbacks import EarlyStopping
from sklearn.preprocessing import OneHotEncoder
from keras.optimizers import Adam
from keras.utils.np_utils import to_categorical

main = Tk()
main.title("Fake Account Detection Using Machine Learning and Data Science")
```

```
main.geometry("1300x1200")
main.config(bg="lightgreen")

global filename
global X, Y
global X_train, X_test, y_train, y_test
global accuracy
global dataset
global model



def loadProfileDataset():
    global filename
    global dataset
    outputarea.delete('1.0', END)
    filename = filedialog.askopenfilename(initialdir="Dataset")
    outputarea.insert(END,filename+" loaded\n\n")
    dataset = pd.read_csv(filename)
    outputarea.insert(END,str(dataset.head()))

def preprocessDataset():
    global X, Y
    global dataset
    global X_train, X_test, y_train, y_test
    outputarea.delete('1.0', END)
    X = dataset.values[:, 0:8]
    Y = dataset.values[:, 8]
    indices = np.arange(X.shape[0])
    np.random.shuffle(indices)
    X = X[indices]
    Y = Y[indices]
    Y = to_categorical(Y)
    X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2)

    outputarea.insert(END,"\n\nDataset contains total Accounts : "+str(len(X))+"\n")
    outputarea.insert(END,"Total profiles used to train ANN algorithm : "+str(len(X_train))+"\n")
    outputarea.insert(END,"Total profiles used to test ANN algorithm  : "+str(len(X_test))+"\n")

def executeANN():
```

```
    global model
    outputarea.delete('1.0', END)
    global X_train, X_test, y_train, y_test
    global accuracy

    model = Sequential()
    model.add(Dense(200, input_shape=(8,), activation='relu', name='fc1'))
    model.add(Dense(200, activation='relu', name='fc2'))
    model.add(Dense(2, activation='softmax', name='output'))
    optimizer = Adam(lr=0.001)
    model.compile(optimizer, loss='categorical_crossentropy', metrics=['accuracy'])
    print('ANN Neural Network Model Summary: ')
    print(model.summary())
    hist = model.fit(X_train, y_train, verbose=2, batch_size=5, epochs=200)
    results = model.evaluate(X_test, y_test)
    ann_acc = results[1] * 100
    print(ann_acc)
    accuracy = hist.history
    acc = accuracy['accuracy']
    acc = acc[199] * 100
    outputarea.insert(END,"ANN model generated and its prediction accuracy is : "+str(acc)+"\n")


def graph():
    global accuracy
    acc = accuracy['accuracy']
    loss = accuracy['loss']



    plt.figure(figsize=(10,6))
    plt.grid(True)
    plt.xlabel('Iterations')
    plt.ylabel('Accuracy/Loss')
    plt.plot(acc, 'ro-', color = 'green')
    plt.plot(loss, 'ro-', color = 'blue')
    plt.legend(['Accuracy', 'Loss'], loc='upper left')
    #plt.xticks(wordloss.index)
    plt.title('ANN Iteration Wise Accuracy & Loss Graph')
    plt.show()
```

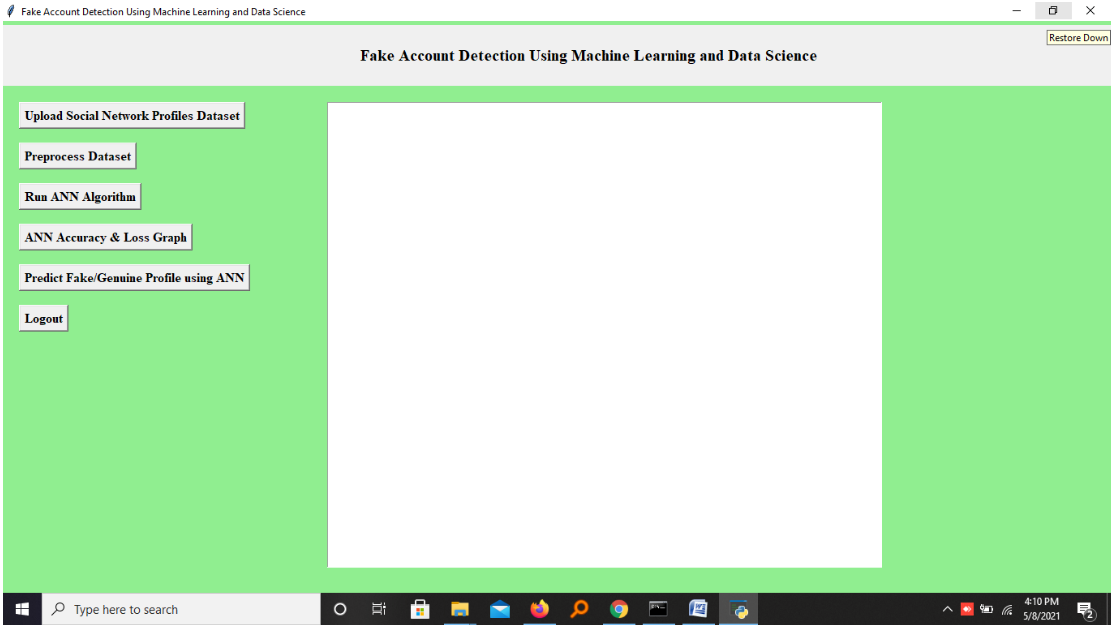
```
def predictProfile():
    outputarea.delete('1.0', END)
    global model
    filename = filedialog.askopenfilename(initialdir="Dataset")
    test = pd.read_csv(filename)
    test = test.values[:, 0:8]
    predict = model.predict_classes(test)
    print(predict)
    for i in range(len(test)):
        msg = ''
        if str(predict[i]) == '0':
            msg = "Given Account Details Predicted As Genuine"
        if str(predict[i]) == '1':
            msg = "Given Account Details Predicted As Fake"
        outputarea.insert(END,str(test[i])+" "+msg+"\n\n")

def close():
    main.destroy()

font = ('times', 15, 'bold')
title = Label(main, text='Fake Account Detection Using Machine Learning and Data Science')
#title.config(bg='powder blue', fg='olive drab')
title.config(font=font)
title.config(height=3, width=120)
title.place(x=0,y=5)

font1 = ('times', 13, 'bold')
ff = ('times', 12, 'bold')

uploadButton = Button(main, text="Upload Social Network Profiles Dataset",
command=loadProfileDataset)
uploadButton.place(x=20,y=100)
uploadButton.config(font=ff)



processButton = Button(main, text="Preprocess Dataset", command=preprocessDataset)
processButton.place(x=20,y=150)
processButton.config(font=ff)

annButton = Button(main, text="Run ANN Algorithm", command=executeANN)
```

```
annButton.place(x=20,y=200)
annButton.config(font=ff)

graphButton = Button(main, text="ANN Accuracy & Loss Graph", command=graph)
graphButton.place(x=20,y=250)
graphButton.config(font=ff)

predictButton = Button(main, text="Predict Fake/Genuine Profile using ANN",
command=predictProfile)
predictButton.place(x=20,y=300)
predictButton.config(font=ff)

exitButton = Button(main, text="Logout", command=close)
exitButton.place(x=20,y=350)
exitButton.config(font=ff)


font1 = ('times', 12, 'bold')
outputarea = Text(main,height=30,width=85)
scroll = Scrollbar(outputarea)
outputarea.configure(yscrollcommand=scroll.set)
outputarea.place(x=400,y=100)
outputarea.config(font=font1)

main.config()
main.mainloop()
```

Chapter 5

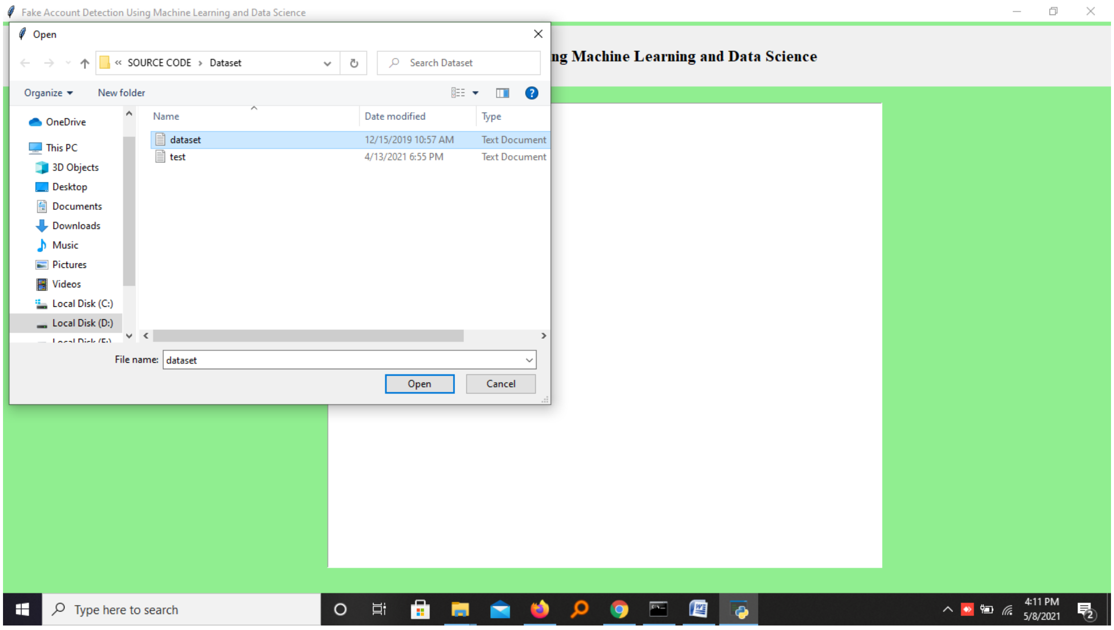
SCREENSHOTS

5.1 Introduction



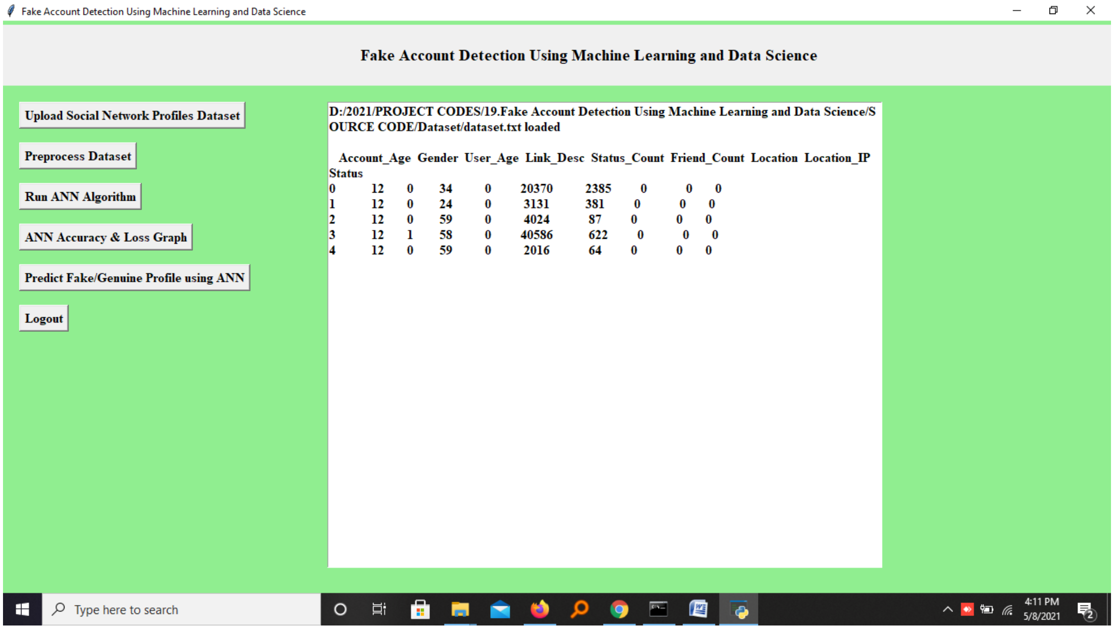
Screenshot 5.1: name of screen

In above screen click on ‘Upload Social Network Profiles Dataset’ button and upload dataset



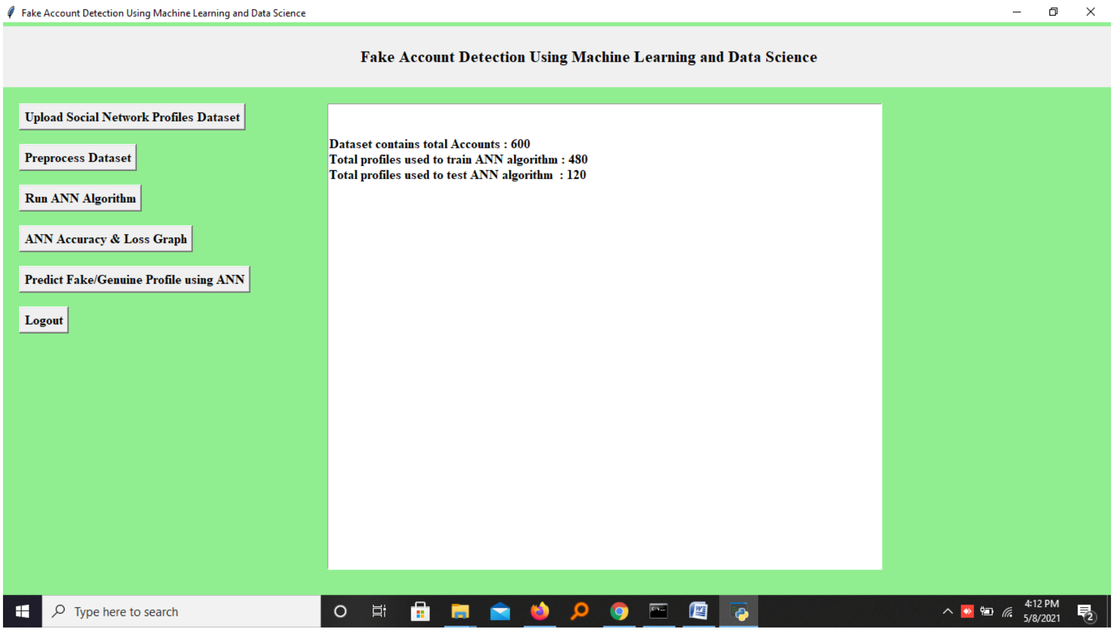
Screenshot 5.2: name of screen

In above screen selecting and uploading ‘dataset.txt’ file and then click on ‘Open’ button to load dataset and to get below screen



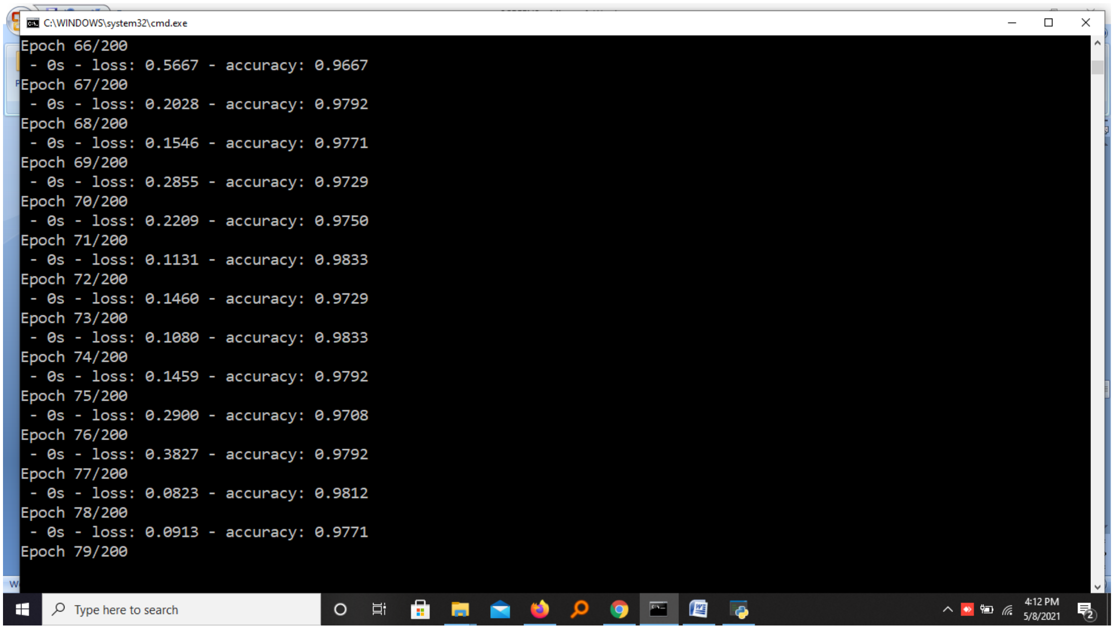
Screenshot 5.3: name of screen

In above screen dataset loaded and displaying few records from dataset and now click on 'Preprocess Dataset' button to remove missing values and to split dataset into train and test part.



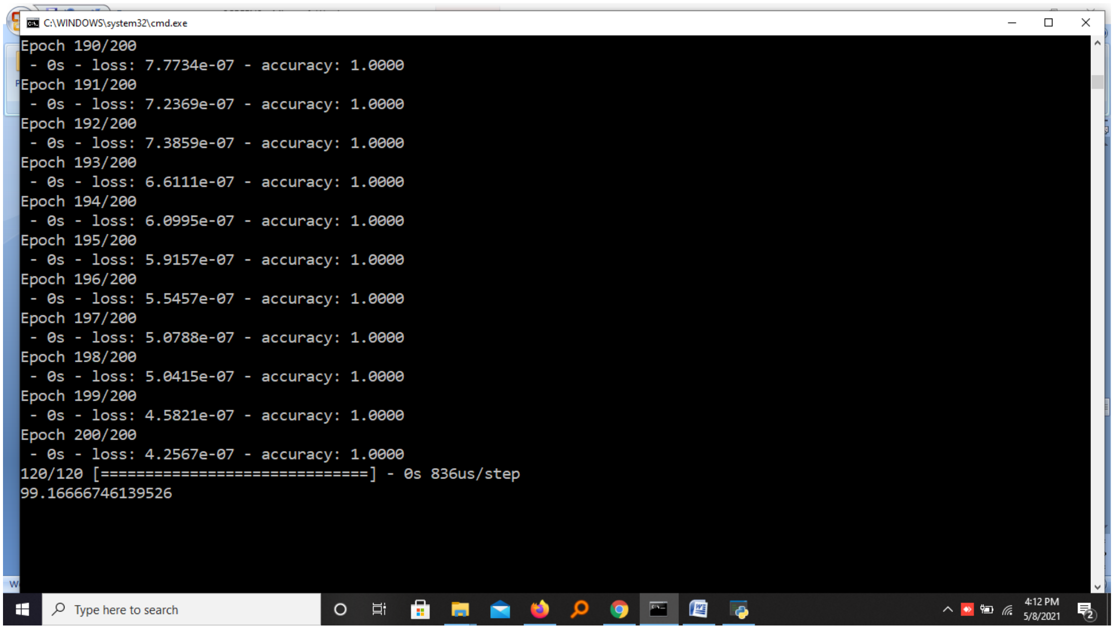
Screenshot 5.4: name of screen

In above screen we can see dataset contains total 600 records and application using 480 records for training and 120 records to test ANN and now dataset is ready and now click on 'Run ANN Algorithm' button to ANN algorithm



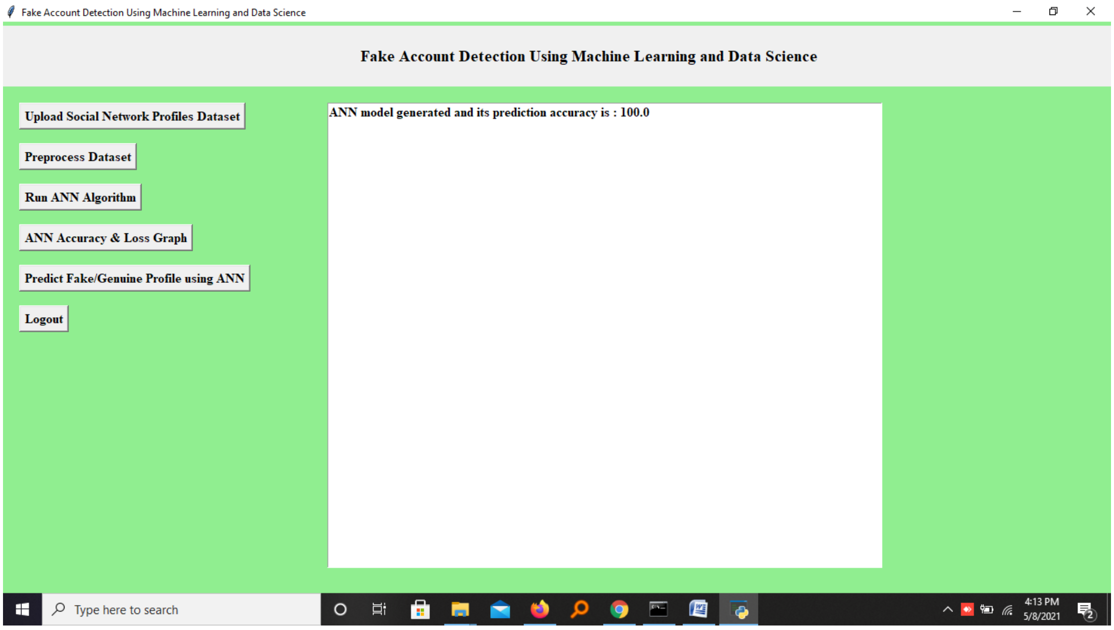
Screenshot 5.5: name of screen

In above screen we can see ANN start iterating model generation and at each increasing epoch we can see accuracy is getting increase and loss getting decrease.



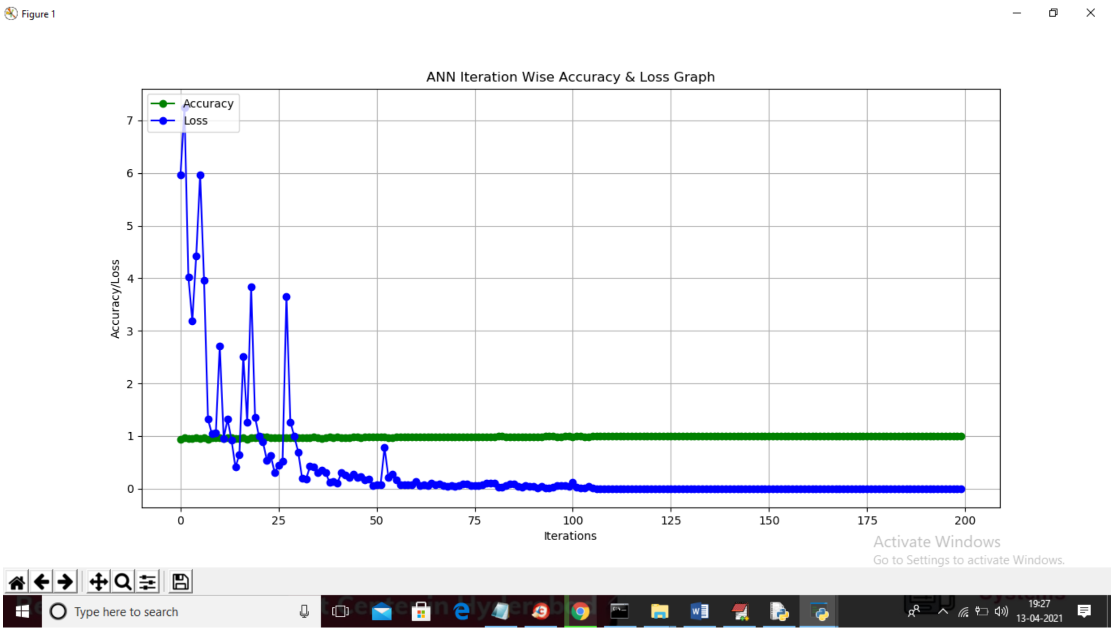
Screenshot 5.6: name of screen

In above screen we can see after 200 epoch ANN got 100% accuracy and in below screen we can see final ANN accuracy



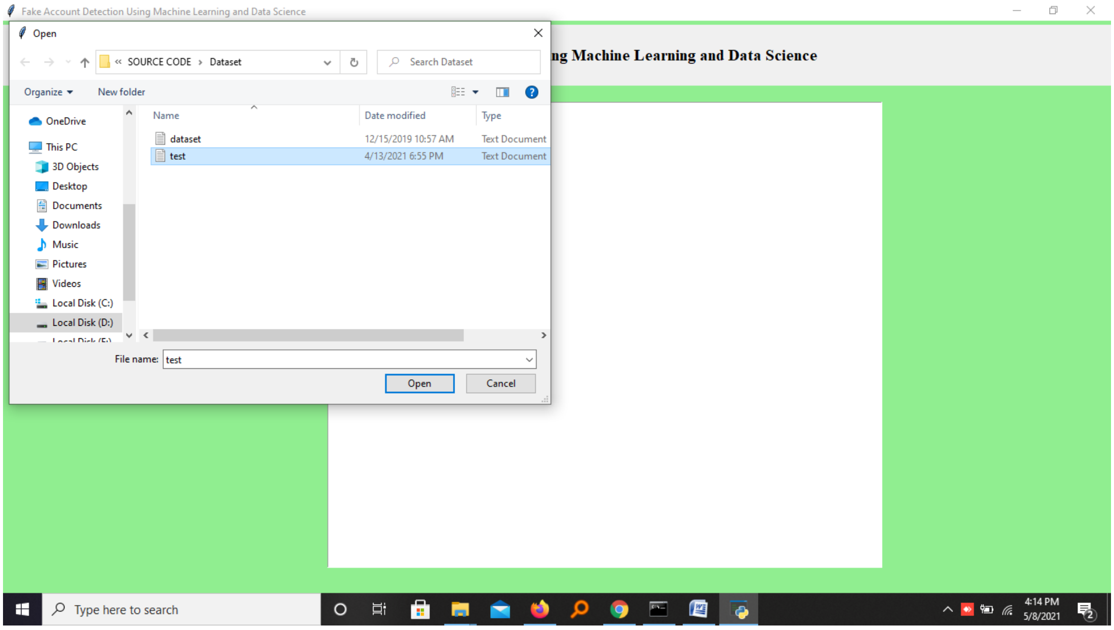
Screenshot 5.7: name of screen

In above screen ANN model generated and now click on 'ANN Accuracy & Loss Graph' button to get below graph



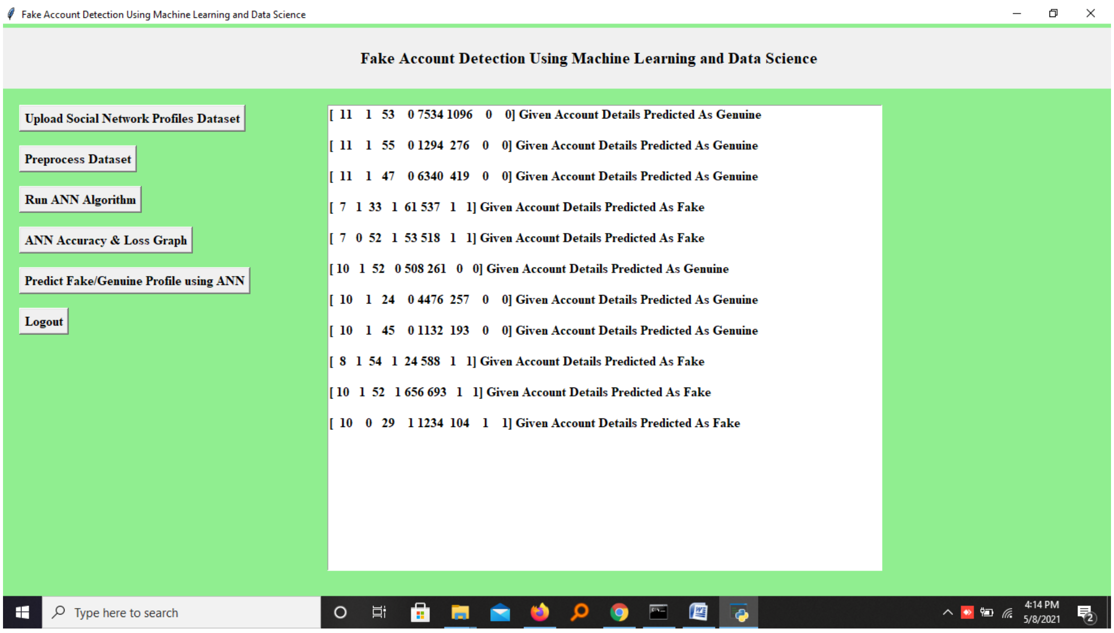
Screenshot 5.8: name of screen

In above graph x-axis represents epoch and y-axis represents accuracy/loss value and in above graph green line represents accuracy and blue line represents loss value and we can see accuracy was increase from 0.90 to 1 and loss value decrease from 7 to 0.1. Now model is ready and now click on 'Predict Fake/Genuine Profile using ANN' button to upload test data and then ANN will predict below result



Screenshot 5.9: name of screen

In above screen we are selecting and uploading 'test.txt' file and then click on 'Open' button to load test data and to get below prediction result



Screenshot 5.10: name of screen

In above screen in square bracket we can see uploaded test data and after square bracket we can see ANN prediction result as genuine or fake

Chapter 6
TESTING

6.1 Introduction to Testing

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

6.2 Types of Testing

6.2.1 Unit Testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

6.2.2 Integration Testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

6.2.3 Functional Testing

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

6.3 Test Cases

S NO	Category	Description	Input	Expected Output
1	Data Preprocessing	Handle missing values	Dataset with missing values	Values imputed/removed
2	Data Preprocessing	Validate normalization	Raw feature values	Scaled/normalized data
3	Data Preprocessing	Check duplicate records	Dataset with duplicates	Duplicates removed
4	Feature Engineering	Feature selection	Raw dataset	Relevant features extracted
5	Feature Engineering	Feature transformation	Categorical & numerical data	Encoded categorical features
6	Feature Engineering	Feature scaling	Feature set	Scaled values
7	Model Training	Verify training process	Training dataset	Model trained successfully
8	Model Training	Hyper parameter tuning	Multiple configuration s	Best configuration selected
9	Model Training	Over fitting prevention	Training validation sets	Model avoids over fitting
10	Model Evaluation	Accuracy verification	Test dataset	Accuracy meets threshold
11	Model Evaluation	Precision, recall, F1-score check	Test dataset	Scores meet benchmark
12	Model Evaluation	Confusion matrix results	Model predictions	Balanced TP, FP, TN, FN values
13	Fake Account Detection	Detect fake account	Fake account profile	Identified as fake
14	Fake Account Detection	Detect real account	Legitimate account profile	Identified as real
15	Fake Account Detection	Evaluate mixed dataset	Dataset with real & fake	Correct classification
16	API Testing	Validate API response format	API request with user data	Correct JSON response
17	API Testing	Check API response time	API request	Response within limits

18	API Testing	Verify API security	Unauthorized API access	Access denied
19	Performance Testing	Test with large dataset	High-volume data	Model handles efficiently
20	Performance Testing	Measure inference time	Multiple requests	Response within limits
21	Performance Testing	Validate system resource usage	High-load scenario	Acceptable CPU & memory usage

Chapter 7

CONCLUSION & FUTURE SCOPE

7.1. Conclusion

In this research, we have come up with an ingenious way to detect fake accounts on OSNs By using machine learning algorithms to its full extent, we have eliminated the need for manual prediction of a fake account, which needs a lot of human resources and is also a time-consuming process. Existing systems have become obsolete due to the advancement in the creation of fake accounts. The factors that the existing system relayed upon is unstable. In this research, we used stable factors such as engagement rate, artificial activity to increase the accuracy of the prediction.

7.2 Future Scope

- Enhancing Model Performance
- Real-time Detection
- Explainability & Interpretability
- Integration with Big Data Technologies
- Cross-Platform Adaptability
- Enhanced Security Measures
- User Awareness and Reporting Mechanism

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